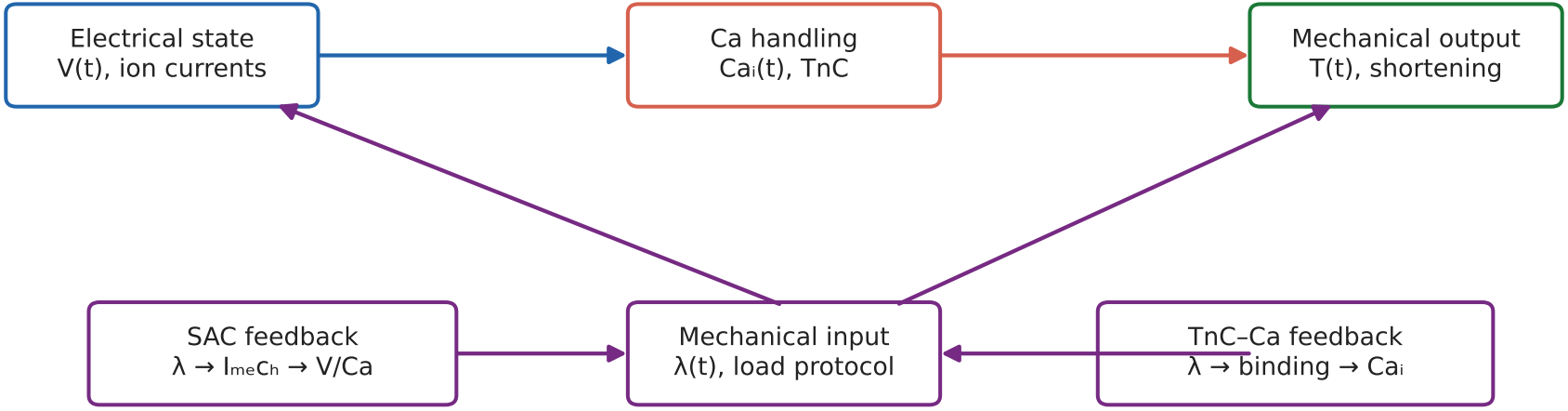
Human bidirectional electromechanical virtual-cell platform



Individualization: baseline + mechanical calibration → cell-specific state → blind counterfactual prediction

Solid arrows: implemented coupling. Red boxes: perturbable mechanical feedback paths.